FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – III

PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C, D**	21.	**A**	41.	**A, B, C**
2.	**A, C**	22.	**B, C, D**	42.	**B, C, D**
3.	**A, D**	23.	**A**	43.	**A, B, C, D**
4.	**A, B, D**	24.	**B**	44.	**A, D**
5.	**A, B**	25.	**A, C, D**	45.	**A, B**
6.	**B, D**	26.	**D**	46.	**B, C**
7.	**C, D**	27.	**A**	47.	**A, B**
8.	**A, D**	28.	**D**	48.	**A, D**
9.	**B, C**	29.	**A, B, C**	49.	**A**
10.	**B, C**	30.	**A, B**	50.	**A, B**
11.	**(A) → (q, s)** **(B) → (p, r)** **(C) → (q, s)** **(D) → (t)**	31.	**A → p** **B → q, s** **C → q, s, t** **D → r, t**	51.	**(A) → (q, t)** **(B) → (p)** **(C) → (s)** **(D) → (r)**
12.	**(A) → (r, s)** **(B) → (r, s)** **(C) → (p)** **(D) → (q, s)**	32.	**A → p, s** **B → p, q** **C → r, t** **D → p, q**	52.	**(A) → (s)** **(B) → (p)** **(C) → (q)** **(D) → (r, t)**
13.	**4**	33.	**1**	53.	**1**
14.	**4**	34.	**9**	54.	**3**
15.	**9**	35.	**3**	55.	**1**
16.	**1**	36.	**4**	56.	**5**
17.	**2**	37.	**5**	57.	**3**
18.	**3**	38.	**5**	58.	**3**
19.	**3**	39.	**5**	59.	**6**
20.	**2**	40.	**5**	60.	**2**

Physics

PART – I

SECTION – A

1. Avg. acceleration $= \dfrac{\text{change in velocity}}{\text{Time interval}}$

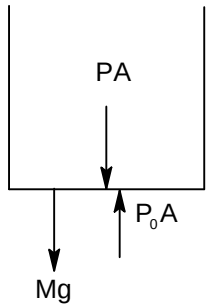
2. Upto time t = g/k, N will be along upwards direction after that it will be along downward direction.

3. $Mg + PA = P_0A$

$$M = \frac{P_0A}{g} - \frac{PA}{g}$$

$$P = \frac{P_i}{2} = \frac{P_0 + \frac{mg}{A}}{2}$$

$$\Rightarrow \quad M = \frac{P_0A}{2g} - \frac{m}{2}$$



4. $B \to C$; $C = C_v + R/2 = 2R$

So; $Q_{B\to C} = nC\,(T_C - T_B) = 2nR(T_C - T_B)$

Also; $\dfrac{T_C}{T_B} = 4, \dfrac{T_A}{T_B} = 2$

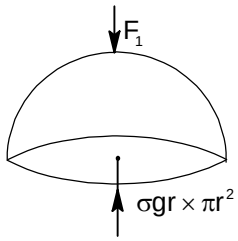
Thus; $Q_{B\to C} = 6nR\,T_B$.

$$Q_{net} = \frac{-nRT_B}{2} \Rightarrow Q_{C\to A\to B} = -130J$$

5. $F_B = \pi r^3\sigma g - F_1 = \dfrac{2}{3}\pi r^3\sigma g$

$$\Rightarrow \quad F_1 = \frac{1}{3}\pi r^3\sigma g$$

Thus; $F_2 = \dfrac{5}{3}\pi r^3\sigma g$



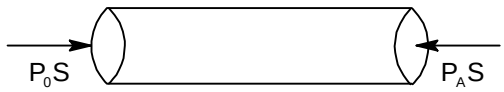
6. By Newton's second law;

$$(p_A - p_0)s = \rho sh\omega^2\left(\ell - \frac{h}{2}\right)$$

$$\Rightarrow \quad p_A = p_0 + \rho\omega^2 h\left(\ell - \frac{h}{2}\right).$$

By Bernoulli's equation;

$$u = \omega\sqrt{2h\left(\ell - \frac{h}{2}\right)}$$



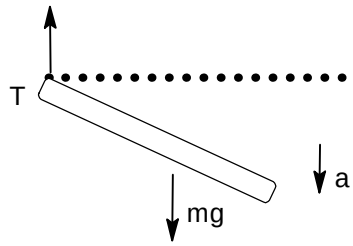
7. $mg - T = ma$

$T\frac{\ell}{2}\cos\theta = \frac{m\ell^2}{12}\alpha$

By constraints;

$a = \frac{\alpha\ell}{2}\cos\theta$

Thus; $T = \frac{mg}{1+3\cos^2\theta}$ and $a_A = \frac{3g\sin\theta\cos\theta}{1+3\cos^2\theta}$



8. $\frac{2mx_B + mx_A}{3m} = \frac{a}{3} \Rightarrow x_A = a - 2x_B$

$\frac{2my_B + my_A}{3m} = 0 \Rightarrow y_A = -2y_B$

Trajectory of B is $\frac{(a-2x)^2}{a^2} + \frac{(2y)^2}{b^2} = 1$

9. $m_B = 2m_A \Rightarrow Z_B = 2Z_A$. Let $Z_A = Z$, $Z_B = 2Z$

So; $E_B - E_A = \frac{13.6\times 5}{36}\times 3Z^2 = 5.667$

$\Rightarrow$ Z = 1

10. $E = m_1\omega_1^2A_1^2 = m_2\omega_2^2A_2^2$

$f = \frac{V}{2L} \Rightarrow \frac{f_1}{f_2} = \frac{\omega_1}{\omega_2} = \frac{L_2}{L_1} = \frac{4}{1}$

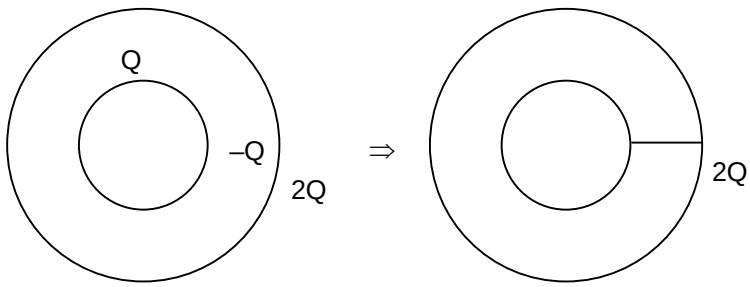
$\frac{A_1}{A_2} = \sqrt{\frac{m_2}{m_1}\times\left(\frac{\omega_2}{\omega_1}\right)^2}$

$= \sqrt{\frac{L_2}{L_1}\times\left(\frac{L_1}{L_2}\right)^2} = \frac{1}{2}$

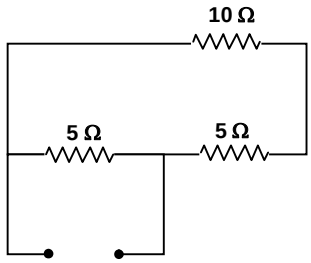
SECTION – C

13. Heat generated = $\frac{Q^2}{2C}$

$= \frac{Q^2}{2\times\frac{4\pi\varepsilon_0}{\frac{1}{a}-\frac{1}{2a}}} = \frac{Q^2}{16\pi\varepsilon_0 a}$



14. $x = \frac{15\times 5}{15+5} = \frac{5}{4}\Omega$



15. $\frac{2}{3}\ell_1 = \frac{3}{4}\ell_2 \Rightarrow \frac{\ell_1}{\ell_2} = \frac{9}{8}$.

$M_1 = 3M$, $M_2 = 4M$.

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{\ell_1}{\ell_2}\right)^3 \times \frac{M_2}{M_1} = \left(\frac{9}{8}\right)^3 \times \frac{4}{3}$$

$$\Rightarrow \quad \frac{T_1}{T_2} = \frac{9}{8}\sqrt{\frac{3}{2}}$$

16. Required mass = $\frac{10^9}{Q-\text{value}\times 0.5} \times 2 \times m\left({}^2_1D\right) \times 24 \times 3600 = 1.3 \text{ kg}$

17. Amplitude of block = $\frac{mg}{k} = 1$ mm

$$\frac{1}{v} + \left(\frac{1}{-30}\right) = \frac{-1}{20}$$

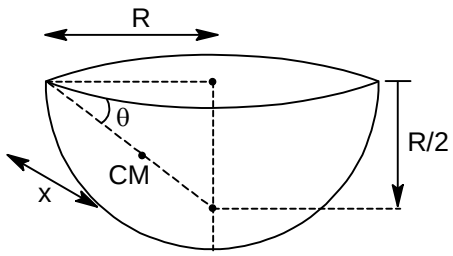
$\Rightarrow v = -60$ cm

Magnification, $m = -\frac{v}{u} = -2$

This; amplitude of image = 2mm.

18. $V_y = \omega x \cos\theta = \frac{\omega R}{2}$

Using conservation of angular momentum about centre of mass.



$$mu\frac{R}{4} = \left\{\frac{2}{3}mR^2 - \frac{mR^2}{4} + \frac{5mR^2}{16}\right\}\omega + \frac{5mR^2}{16}\omega$$

$$= \frac{25}{24}mR^2\omega$$

$$\Rightarrow \quad \omega = \frac{6u}{25R}$$

$$\Rightarrow \quad V_y = \frac{3u}{25}$$

19. $L\frac{di}{dt} = B\ell v$

$$m\frac{dv}{dt} = -iB\ell$$

$$\Rightarrow \frac{d^2v}{dt^2} = -\left(\frac{B^2\ell^2}{mL}\right)v$$

$\Rightarrow v = v_0 \cos(\omega t)$

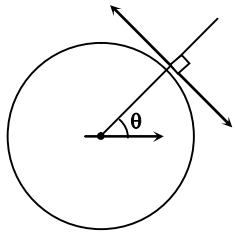
with $\omega = \frac{B\ell}{\sqrt{mL}}$

Thus, $\frac{dx}{dt} = v_0 \cos(\omega t)$

$\Rightarrow x = \frac{v_0}{\omega}\sin\omega t$

$2A = \frac{2v_0}{\omega} = 3$ cm

20. Tangential components must cancel.



$\Rightarrow \frac{\sigma}{2\varepsilon_0}\sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{R^3}$

$\Rightarrow R^3 = \frac{p}{2\pi\sigma}$

$\Rightarrow n = 2$

Chemistry

PART – II

SECTION – A

21. $M_{H_2SO_4} = \frac{24.7}{98} \times \frac{1000 \times 1.18}{100} = 2.97$

$N_{H_2SO_4} = 5.94$

Meq. of H_2SO_4 = meq. of NaOH + meq. of Al

$5.94 \times 75 = 0.5 \times V_{NaOH} + 300$

$V_{NaOH} = 291$ ml

22. According to Charle's law $V \propto T \Rightarrow V = KT$

$$\left(\frac{dV}{dT}\right)_P = K$$

$$\left(\frac{dT}{dV}\right)_P = K$$

$$\frac{d}{dT}\left(\frac{V}{T} - K\right)_P = 0$$

$$\left(\frac{1}{T} - \frac{V}{T^2}\right)_P = 0$$

23. Volume of unit cell = 1.25×10^{-22} cc

Mass of unit cell = $1.25 \times 10^{-22} \times 4$

$= 5 \times 10^{-22}$ gm

For one molecule $= \frac{72}{6.022 \times 10^{23}} = 1.195 \times 10^{-22}$ g

Hence number of FeO molecules per unit cell $= \frac{5 \times 10^{-22}}{1.195 \times 10^{-22}} = 4.19 \approx 4$

Hence there are $4Fe^{2+}$ and $4O^{2-}$ in each unit cell.

24. $$h\nu_1 = h\nu_o - \frac{1}{2}mu_1^2 \quad \ldots (1)$$

$$h\nu_2 = h\nu_o - \frac{1}{2}mu_2^2 \quad \ldots (2)$$

$$\frac{1}{2}mu_1^2 = \frac{1}{K}\left(\frac{1}{2}mu_2^2\right)$$

From equation (1)

$$h\nu_1 = h\nu_o - \frac{1}{2k}mu_2^2 \quad \ldots (3)$$

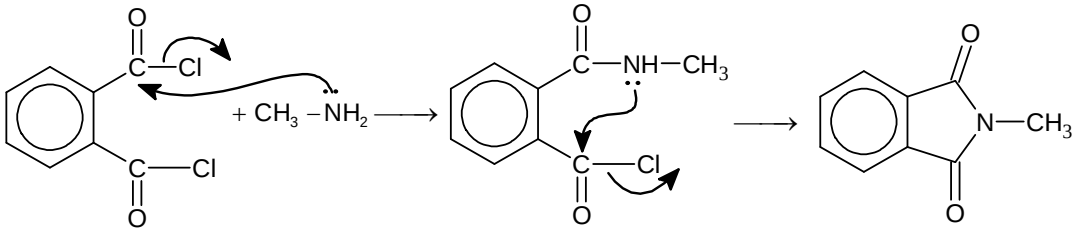
$$\frac{1}{2}mu_2^2 = Kh\nu_o - Kh\nu_1 \quad \ldots (4)$$

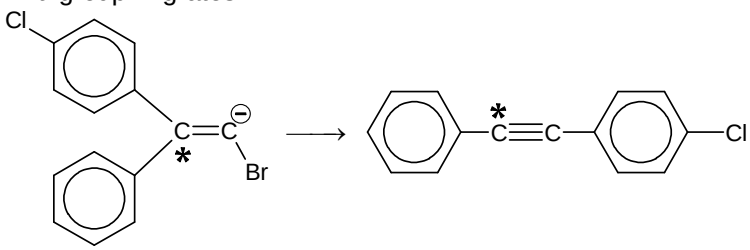
From Eqn. (4) & (2)

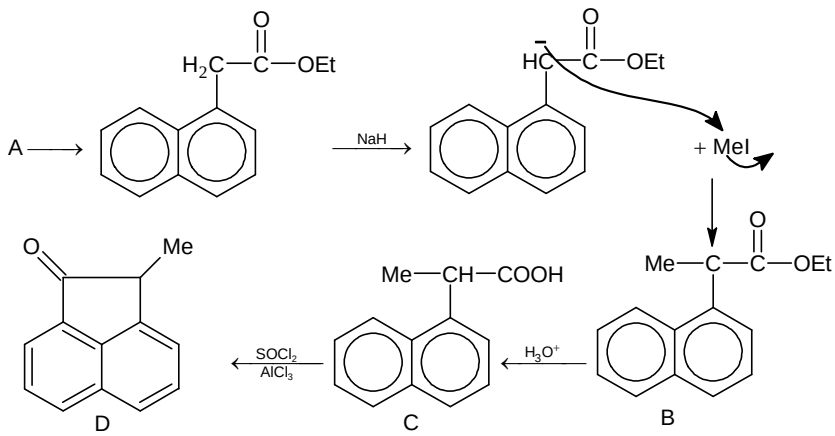
$$h\nu_2 = h\nu_o - Kh\nu_o + Kh\nu_1$$

$$\nu_o = \frac{K\nu_1 - \nu_2}{(K-1)}$$

25. Work is a form of exchange energy between system and surrounding in adiabatic process.
- Intensive property is not additive.
- In free expansion work done is zero.
- For an isolated system the entropy either increases or remains constant.

26.


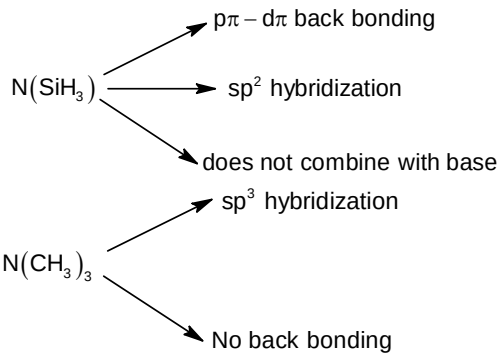
27. Anti group migrates


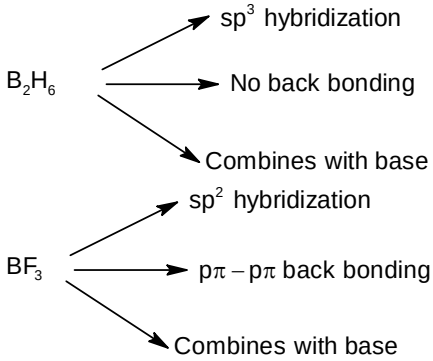
28.


29. Fact based question.

30. Fact based question.

SECTION-B

31.




32. $[Ma_3b_2c]$ Geometrical isomerism → 3 Stereoisomer → 3

$[Ma_3b_3]$ Geometrical isomerism → 2 Stereoisomer → 2

$[Ma_3bcd]$ Geometrical isomerism → 4 Stereoisomer → 5

$[Ma_4bc]$ Geometrical isomerism → 2 Stereoisomer → 2

SECTION – C

33. $\underset{30}{\underset{0}{Cl_2}} + \underset{30}{\underset{0}{Cl_2O_n}} \longrightarrow \underset{0}{\underset{60}{2Cl_2}} + \underset{0}{\underset{15n}{\frac{n}{2}O_2}}$

KOH absorbs Cl_2

So, 15n = 15

n = 1

34. $27.2 \times 1.602 \times 10^{-12} = R_H z^2 hc\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$ … (1)

$10.2 \times 1.602 \times 10^{-12} = R_H z^2 hc\left[\frac{1}{3^2} - \frac{1}{n^2}\right]$ … (2)

After solving (1) and (2)

n = 6, z = 3

n + z = 9

35. Specific heat × At. Wt. = 6.4

Atomic weight = 29.63

Eq. of metal = Eq. of Cl_2

$\frac{7.467 \times x}{29.63} = \frac{71 \times 9.2}{22.4 \times 35.5}$

x = 3.25

x = 3 (x is an integer)

36. $\Delta T = \frac{1000 \times K_b' \times w}{m \times W}$

m_{exp} = 123.8

$\frac{m_N}{m_{exp}} = 1 - \alpha + \frac{\alpha}{n}$

$\because \alpha = 1 \Rightarrow \frac{m_N}{m_{exp}} = \frac{\alpha}{n}$

$\frac{31}{123.8} = \frac{1}{n} \Rightarrow n \approx 4$

37. $Fe^{3+} + H_2S \longrightarrow S + Fe^{2+}$

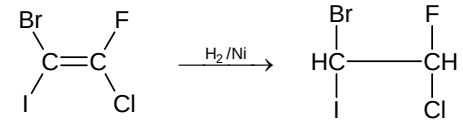
$SO_2 + 2H_2S \longrightarrow 3S + 2H_2O$

$H_2O_2 + H_2S \longrightarrow 2H_2O + S$

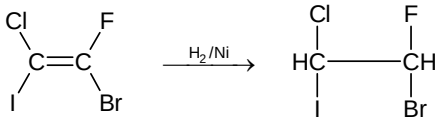
$H_2S + O_3 \longrightarrow H_2O + S + O_2$

$2HNO_3 + H_2S \longrightarrow 2NO_2 + S + 2H_2O$

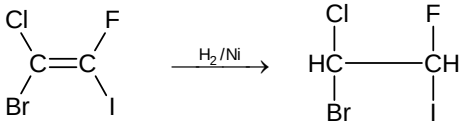
38. All nitrates of alkali earth metal decompose like this.

39. 

E (or) z give same product.

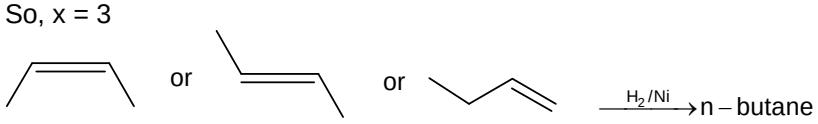


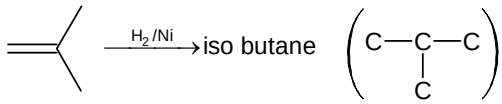
E (or) z give same product.



E (or) z give same product.

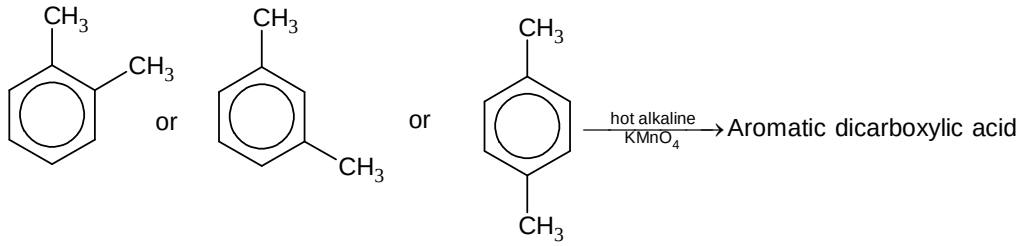
So, x = 3



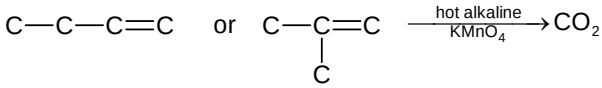


Y = 2

$\therefore X + Y = 5$

40. 

So X = 3



Y = 2

X + Y = 5

Mathematics

PART – III

SECTION – A

41. $$\sum_{1\le i<j\le 7}\left|a_i+a_j\right|^2=\sum_{1\le i<j\le 7}\left(|a_i|^2+|a_j|^2+a_i\bar{a}_j+\bar{a}_i a_j\right)$$

$$=2R^2\cdot{}^7C_2+\sum_{i\neq j}a_i\bar{a}_j=42R^2+\sum_{i=1}^{7}\sum_{j=1}^{7}a_i\bar{a}_j-\sum_{i=1}^{7}a_i\bar{a}_i$$

$$=35R^2+\left(\sum_{i=1}^{7}a_i\right)\left(\sum_{i=1}^{7}\bar{a}_i\right)-7R^2$$

$$=35R^2+\left|\sum_{i=1}^{7}a_i\right|^2\ge 35R^2$$

42. $HA_1'=-2R\cos A_7\cos A_9$

$HA_7'=-2R\cos A_1\cos A_7$

$HA_9'=-2R\cos A_1\cos A_7$

$\angle A_1=\frac{\pi}{13}$, $\angle A_7=\frac{9\pi}{13}$, $\angle A_9=\frac{3\pi}{13}$ so on using, we get $HA_1'-HA_7'+HA_9'=\frac{R}{2}$

43. $$\left|\int_0^a f(x)\,dx\right|=\left|a\int_0^1 f(at)\,dt\right|\le\frac{1}{8}M$$

44. $$\begin{vmatrix}x & y & z\\ 2 & -3 & 1\\ 1 & 1 & 1\end{vmatrix}=0;\quad\begin{vmatrix}x-2 & y-1 & z+2\\ 3 & -5 & 2\\ 1 & 1 & 1\end{vmatrix}=0$$

45. $\vec{p}\times(\vec{x}\times\vec{q})=(\vec{p}\cdot\vec{q})\vec{x}-(\vec{p}\cdot\vec{x})\vec{q}=\vec{p}\times\vec{r}$

$$\vec{x}=\frac{(\vec{p}\cdot\vec{x})\vec{q}}{\vec{p}\cdot\vec{q}}+\frac{(\vec{p}\times\vec{r})}{\vec{p}\cdot\vec{q}}=\frac{\alpha}{\vec{p}\cdot\vec{q}}\vec{q}+\frac{(\vec{p}\times\vec{r})}{\vec{p}\cdot\vec{q}}$$

46. Property

47. $$1+{}^4C_1\times 2+\frac{\lfloor 4}{\lfloor 2\lfloor 2\lfloor 2}\times\lfloor 2+\frac{\lfloor 4\times\lfloor 3}{\lfloor 2\lfloor 1\lfloor 1\lfloor 2}+\lfloor 4=75$$

48. Using property and simplifying

49. Odd and even object draw is a bijection

$\therefore\ \frac{1}{2}$

50. $g(x)=(x+1)f(x)-x$

$g(x)=0$ for $x=0, 1, 2, \ldots\ldots n$

$g(x)=a\,x(x-1)\ldots\ldots(x-n)$

$$\therefore \quad \frac{f(x) = \frac{(-1)^{n+1} x.(x-1).....(x-n)}{\lfloor n+1} + x}{x+1}$$

$$\Rightarrow f(n+1) = \begin{cases} 1 & , \text{ odd 'n'} \\ \frac{n}{n+2} & , \text{ even 'n'} \end{cases}$$

SECTION – B

51. Plane through line y = ax, z = c is $(y - ax) + \lambda_1(z - c) = 0$
Similarly, $(y + ax) + \lambda_2(z + c) = 0$
$\Rightarrow (y + c) + \lambda_1(y - c) = 0$ and $(y - c) + \lambda_2(y + c) = 0$
$\Rightarrow -\lambda_1 = \frac{y+c}{y-c}$, $\frac{-1}{\lambda_2} = \frac{y+c}{y-c}$; $\lambda_1\lambda_2 = 1$
$\therefore$ Locus is $\left(\frac{y-ax}{z-c}\right)\left(\frac{y+ax}{z+c}\right) = 1$
$a^2x^2 - y^2 + z^2 - c^2 = 0$

52. KL = 2R sin A
LM = 2R sin B
MK = 2R sin C
Δ KLM ~ ΔABC
$\Rightarrow$ Ar(ΔAKM) = Ar(ΔBLK) = Ar(CLM) = $\frac{2}{9}$Ar(ΔABC)
$\Rightarrow \text{Ar}(\Delta\text{KLM}) = \frac{1}{3}\text{Ar}(\Delta\text{ABC})$
$\therefore a^2 = b^2 + c^2 - 2bc\cos A$
$\Rightarrow \frac{1}{3}a^2 = \left(\frac{2b}{3}\right)^2 + \left(\frac{c}{3}\right)^2 - 2\cdot\frac{2b}{3}\cdot\frac{c}{3}\cos A$
$\Rightarrow a^2 = 2b^2 - c^2$
Similarly, $b^2 = 2c^2 - a^2$ and $c^2 = 2a^2 - b^2$
$\Rightarrow a^2 = b^2 = c^2$
$\therefore a = b = c$

SECTION – C

53. Using property of matrix and expanding

54. A(a, 0), B(0, b)
Put $y = \frac{1}{x}$
$x^2 - ax + \frac{a}{b} = 0$ and $x = \frac{1}{y}$
$y^2 - by + \frac{b}{a} = 0$
$AP^2 = (x_1 - a)^2 + y_1^2 = (a - x_2 - a)^2 + (b - y_2)^2 = x_2^2 + (b - y_2)^2 = BQ^2$

55. $\beta = \dfrac{1}{\alpha\gamma}$

Assume, $\alpha \geq \beta \geq \gamma$ (Say)

$\Rightarrow \alpha \geq 1$ and $\gamma \leq 1$

$\therefore \dfrac{\alpha^{\beta+\gamma} \cdot \gamma^{\alpha+\beta}}{(\alpha\gamma)^{\alpha+\gamma}} = \dfrac{\gamma^{\beta-\gamma}}{\alpha^{\alpha-\beta}} \leq 1$

56. $\lim\limits_{n\to\infty} a_n = a + \dfrac{b-a}{2} + \dfrac{b-a}{8} + ... = a + \dfrac{2}{3}(b-a) = \dfrac{a+2b}{3}$

57. Assume $\alpha_i < \dfrac{\pi}{6}$

$2(\sin\alpha_1 + \sin\alpha_2 + \sin\alpha_3) = 3\sin(\alpha_1 + \alpha_2 + \alpha_3)$

Also, $\dfrac{\sin\alpha_1 + \sin\alpha_2 + \sin\alpha_3}{3} \leq \sin\left(\dfrac{\alpha_1 + \alpha_2 + \alpha_3}{3}\right) \left[\alpha_i \in \left(0, \dfrac{\pi}{2}\right)\right]$

$\Rightarrow \dfrac{\sin(\alpha_1 + \alpha_2 + \alpha_3)}{2} \leq \sin\left(\dfrac{\alpha_1 + \alpha_2 + \alpha_3}{3}\right)$

$\Rightarrow \sin 3t \leq 2 \sin t$

$\Rightarrow 4\sin^3 t - \sin t \geq 0$

$\Rightarrow \sin^2 t \geq \dfrac{1}{4}$, which contradicts assumption

58. It's a sure event so, p = 1

59. $I = \lim\limits_{n\to\infty} \dfrac{\pi}{n}\left[\ln\left(a^2 - 2a\cos\dfrac{\pi}{n} + 1\right) + \ln\left(a^2 - 2a\cos\dfrac{\pi}{n} + 1\right)..... + \ln\left(a^2 - 2a\cos\dfrac{(n-1)\pi}{n} + 1\right)\right]$

$= \lim\limits_{n\to\infty} \dfrac{\pi}{n}\ln\left(\left(a^2 - 2a\cos\dfrac{\pi}{n} + 1\right)\left(a^2 - 2a\cos\dfrac{2\pi}{n} + 1\right).....\left(a^2 - 2a\cos\dfrac{(n-1)\pi}{n}\right) + 1\right)$

$= \lim\limits_{n\to\infty} \prod\limits_{k=1}^{n-1}\left(\left(a - e^{\frac{ik\pi}{n}}\right)\left(a - e^{\frac{-ik\pi}{n}}\right)\right) = \lim\limits_{n\to\infty} \prod\left(\dfrac{a^{2n} - 1}{a^2 - 1}\right)^{\frac{1}{n}} = 2\pi\ln a$

60. $\int\limits_0^1 \left(f(x)\cdot x - (f(x))^2\right)dx = \int\limits_0^1 \dfrac{x^2}{4}dx$

$\int\limits_0^1 \left[(f(x))^2 - x\cdot f(x) + \dfrac{x^2}{4}\right]dx = 0$

$\int\limits_0^1 \left(f(x) - \dfrac{x}{2}\right)^2 dx = 0$

$\Rightarrow f(x) = \dfrac{x}{2}$

$\therefore f'(x) = \dfrac{1}{2}$